

OPINION OF THE COURT

GARCIA, J.

Courts, including the U.S. Supreme Court (*see Williams v Illinois*, 567 US 50 [2012]), have struggled to identify the point at which forensic DNA science and the rights provided by the Sixth Amendment's Confrontation Clause intersect—that is, when the mechanical processing of data becomes testimony.

Recently this Court, in *People v John*, *People v Austin*, and *People v Tsintzelis*, identified as testimonial the editing stage at which “the generated DNA profile” is created by the analyst exercising independent judgment (27 NY3d 294, 310 [2016]; 30 NY3d 98 [2017]; 35 NY3d 925 [2020]). This case requires us to apply this approach and, in so doing, decide whether the record here established that the proper witness testified about the creation of the DNA profiles. Because the record was insufficient, we now reverse.

A store in Queens was robbed, and an employee found a cellphone on the floor that had been left behind when the robbers fled. Police officers swabbed the phone for DNA, and analysts from the Office of Chief Medical Examiner (OCME) created a DNA profile from the swab and ran the profile through the Combined DNA Index System (CODIS). After CODIS identified defendant as the source of the DNA based on a preexisting DNA profile stored in the data bank, she was arrested and charged with second-degree robbery and petit larceny. OCME analysts created a DNA profile from a post-arrest buccal swab sample from defendant and confirmed the match.

At trial, the People called an OCME level 4 criminalist to testify that defendant’s DNA profile matched the profile developed from the crime scene swab. Three relevant exhibits were entered into evidence: a forensic biology file containing all paperwork developed by OCME, including a DNA profile generated from the DNA swabbed from the cellphone; a similar forensic biology file that contained a DNA profile generated from the confirmatory swab taken from defendant; and a demonstrative chart used to explain the matches to the jury. Defense counsel objected, arguing that because the witness did not perform any of the DNA testing himself, defendant’s right to confrontation would be violated by his testimony regarding the DNA profiles. The People responded that the witness had performed an independent analysis of the profiles and so was the proper witness. The court allowed the testimony and admitted all three exhibits. The witness testified about the process for developing DNA profiles and, as discussed in more detail below, described his work on the profiles in this case. With the use of the comparison chart, he testified that “the mixture found on the swab of the entire phone was approximately 25.7 quadrillion [times] more probable if it originated from Donna Jordan and from one unknown person rather than two un-

known persons.” Defendant testified and denied involvement in the robbery.

Following the prosecutor’s summation, defense counsel twice moved for a mistrial on the grounds that the prosecutor’s statements amounted to misconduct. While acknowledging that some of the prosecutor’s statements on summation were inappropriate, the trial court held that the prosecutor’s comments did not rise to the level of prosecutorial misconduct. The jury found defendant guilty on both counts.

The Appellate Division affirmed, holding that defendant’s right to confrontation was not violated because the criminalist “performed his own analysis of the DNA profiles” and that, “[t]o the extent that some of the prosecutor’s remarks were improper, those remarks were not so flagrant or pervasive as to have deprived the defendant of a fair trial” (201 AD3d 946, 948 [2d Dept 2022]). A Judge of this Court granted leave (38 NY3d 1134 [2022]), and we now reverse based on the violation of defendant’s rights under the Confrontation Clause.

It is well settled that “if an out-of-court statement is testimonial in nature, it may not be introduced against the accused at trial unless the witness who made the statement is unavailable and the accused has had a prior opportunity to confront that witness” (*Bullcoming v New Mexico*, 564 US 647, 657 [2011]).¹ In *People v John*, absent controlling U.S. Supreme Court precedent addressing the interplay of the introduction of DNA forensic evidence and the restraints posed by the Confrontation Clause, this Court sought to identify the point at which the processes involved in DNA testing turn from ministerial to testimonial.

We first made clear that the testifying criminalist need not have been involved in each step of the procedures required to generate the DNA profiles (27 NY3d at 312-313). This rule is consistent with the view of other courts that the preliminary steps involved in the creation of DNA samples are “essentially ministerial tasks” (see *United States v Robinson*, 2023 WL 6066691, *4, 2023 US Dist LEXIS 165537, *13-14 [WD Wash, Sept. 18, 2023, No. 2:22-cr-00212-TL, Lin, J.]; see also *Williams*, 567 US at 100-101 [Breyer, J., concurring] [categorizing preliminary stages of DNA testing as “highly automated” and

1. The People’s contention on appeal that the DNA profile from the cellphone swab was not testimonial because no suspect had been identified is unpreserved.

“mostly automated”]). These preliminary stages are processes that simply prepare the DNA samples for testing, during which “the analyst’s role is to facilitate the operation of a machine” (*State v Lui*, 179 Wash 2d 457, 487, 315 P3d 493, 507 [2014 en banc]; see also *John*, 27 NY3d at 300 [describing these procedures as “extraction (to release the DNA from any cells), quantitation (to determine how much DNA was present), (and) amplification (to make millions of copies of the specific locations . . . to be tested)”). Errors made at this stage “may often be detected from the profile itself,” which provides “a further safeguard” (*Williams*, 567 US at 85).

Instead of imposing a rule that required production of every person who comes into contact with the evidence at these preliminary stages of testing, we held instead that the relevant witness is one directly involved in the critical final stage of testing, when an analyst must exercise judgment to cull the data and generate the DNA profile. It is at this later stage of the testing, only after the “output[of] an electropherogram, or a plot of the peaks and valleys in the DNA[, that] any element of human decision-making enter[s] the process; an expert must translate the peaks and valleys of the electropherogram into a DNA profile” (*Lui*, 179 Wash 2d at 487-488, 315 P3d at 507 [emphasis omitted]; see *John*, 27 NY3d at 298 [“Experienced analysts convert these numeric identifiers into a DNA profile using . . . the analyst’s independent manual examination (of the data) which involves an editing process”]). We have held that “a single analyst” may testify so long as it is “an analyst who witnessed, performed or supervised the generation of defendant’s DNA profile, or who used his or her independent analysis on the raw data, as opposed to a testifying analyst functioning as a conduit for the conclusions of others” (*John*, 27 NY3d at 312, 314-315). Accordingly, testimony from an analyst is sufficient where the witness has engaged in “the final-level DNA analysis, reviewed the results of the preliminary evidence processing conducted by colleagues, produced the relevant DNA profiles, and expressed her expert opinion’ ” as to the DNA match (see *Robinson*, 2023 WL 6066691, *4, 2023 US Dist LEXIS 165537, *14; *United States v Peshlakai*, 2023 WL 4235671, *11-12, 2023 US Dist LEXIS 111275, *27-32 [D NM, June 28, 2023, No. 21-cr-01501-JCH, Herrera, J.] [reviewing the stages of DNA testing and holding that no Confrontation Clause violation occurred because the testifying examiner “performed the final analysis, reviewed the final lab-

oratory notes and data, and drew the relevant conclusions”]; *State v Lopez*, 45 A3d 1, 13 [RI 2012] [testifying examiner’s testimony sufficient where, “(a)fter the first three stages of the DNA testing were completed (by others), he then reviewed the entire case file and confirmed that all protocols were followed properly by examining the other analysts’ notes, their affirmations that protocols were followed, as well as their conclusions” and “personally reviewed and independently analyzed” the data]).

[1] In sum, the testifying analyst must have either participated in or directly supervised this “final” step that generates the DNA profile, or must conduct an “independent analysis” of the data used to do so in a manner that enables replication of the determinations made at that stage in order to verify the profile (*see Tsintzelis*, 35 NY3d at 926; *Austin*, 30 NY3d at 105). The record here fails to establish that the testifying analyst had the requisite involvement with either of the DNA profiles.²

First, the criminalist testified about procedures at the laboratory generally, explaining that “the results and conclusions are entered into a computer by a reporting analyst and then only the technical reviewer comes into their own conclusions if those conclusions agree,” while a “reporting analyst is the first person to go through the completed data and draw their own independent conclusions as to the results,” after which they “draft [an] electronic report that summarizes their conclusions and findings, and they assemble the paper case file that accompanies that report.” He later explained that “[t]here is an analyst who will go through that data using a piece of software . . . [that] removes noise along the graph that we know empirically not to be true data,” and any determination that “a piece of information is likely noise . . . is documented within the data and case file as well,” and “after that is complete, that visual determination and recordkeeping, that’s when the interpreting analyst gets the final data for their interpretation, and the technical reviewer then looks at that same data and makes their interpretation.” He did not explain who performed these tasks for the relevant profiles.

The witness next testified that, for the profile from the cellphone swab, he was the technical reviewer, that he did not

2. Defense counsel’s detailed objection to the entirety of the criminalist’s testimony on confrontation grounds sufficiently preserved a challenge to the admission of all relevant DNA evidence.

prepare the reports and was not present when the reports were made, and that he “review[ed] the data and reports” and conducted his “independent interpretation of the testing data contained in” the paper file he received. He testified that he was the reporting analyst for the post-arrest confirmatory swab and, when asked if he “actually ran the raw data and remanded it to the final data,” he explained that a different analyst performed that task and that, “[a]s the reporting analyst, I receive that data and went through the data[,] . . . received the raw data and I go through it to make sure . . . interpretations are being drawn, and then I draw my own interpretations based on that data.” He responded affirmatively when asked if “someone else ran the raw data, and did the work on it and provided it to you and you reviewed it.”

This record was insufficient. We cannot discern from these statements the witness’s level of involvement in the generation of the DNA profiles at the crucial final stage of testing. Testimony that, for one profile, he received reports and reviewed them, and that, for the other, he “received” the raw data and went through it, but that someone else “ran” the raw data and provided it to him, did not demonstrate that he was the proper witness. The witness did not recount any involvement in the stages of testing that require the exercise of judgment and the opportunity to identify error. This witness’s imprecise and conclusory statements failed to establish adequately that he “used his . . . independent analysis on the raw data” and was not merely “functioning as a conduit for the conclusions of others” (*John*, 27 NY3d at 315).

Without adequate testimony, the People could not establish whether the witness participated in or supervised this final step, or performed an independent analysis of the data through which he would have been able to replicate the determinations, identify any errors committed in the earlier stages of testing, and draw his own conclusions. The People should make a clear record of the stages of the DNA testing in which the testifying analyst was involved, or what manner of independent analysis the testifying analyst performed, ensuring that a witness’s testimony establishes his or her involvement at the necessary stage to avoid a Confrontation Clause error. Courts must be able to confirm that the testifying analyst participated in the critical portion of the testing process or reviewed the data in a meaningful way that enabled independent verification of the accuracy of the DNA profile. Here no such record was made,

and accordingly, admission of the criminalist's testimony and the underlying exhibits was error.

[2] This error was not harmless. The constitutional harmless error standard applies here, requiring a showing that the evidence was overwhelming and that there is no reasonable possibility that the error might have contributed to defendant's conviction (*People v Clyde*, 18 NY3d 145, 153-154 [2011]). The People produced only one witness who identified defendant; that witness's initial physical descriptions of the perpetrator did not match defendant's age or height; and defendant presented medical evidence that suggested she was physically incapable of committing the robbery. Accordingly, defendant is entitled to a new trial. As a result, we do not reach defendant's remaining arguments.

We note that our holding today is made absent clear guidance from the U.S. Supreme Court (*see e.g. State v Watson*, 170 NH 720, 733, 185 A3d 845, 855-856 [2018] ["Because each of the opinions in Williams embraces a different approach to determining whether the use of forensic evidence violates the Confrontation Clause, and there is no narrow rule that would have the support of a majority of the Supreme Court, we agree with other jurisdictions that have concluded that its force, as precedent, is at best unclear" (internal quotation marks, brackets and citation omitted)] [citing cases]; *see also Garlick v Lee*, 1 F4th 122, 133 [2d Cir 2021]). The Supreme Court may revisit the issues in *Williams* in a manner that affects today's holding. Technology may also develop in a way that affects confrontation rights and our holding is limited to the process for forensic DNA analysis as currently conducted.

The order of the Appellate Division should be reversed and a new trial ordered.

Chief Judge WILSON and Judges RIVERA, SINGAS, CANNATARO, TROUTMAN and HALLIGAN concur.

Order reversed and a new trial ordered.
